

Group Actions, Permutations & Introduction to Ring Theory

Master Calibrated Lecture Notes and Comprehensive Mathematical Exposition

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September 22, 2026

These notes provide a unified and rigorous exposition of group actions, conjugation, permutation representations, Cayley's theorem, orbit-stabilizer theory, the class equation, p -groups, conjugacy classes in symmetric groups, and introductory ring theory with matrix ring constructions.

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Part I: Group Actions and Permutation Representations

1 Group Actions: Left and Right Actions

Definition 1.1: Left Group Action

Let G be a group and A a non-empty set. A **left group action** of G on A is a mapping

$$* : G \times A \longrightarrow A, \quad (g, a) \longmapsto g * a$$

satisfying the following two axioms:

- (i) $g_1 * (g_2 * a) = (g_1 g_2) * a \quad \forall g_1, g_2 \in G, \forall a \in A,$
- (ii) $e * a = a \quad \forall a \in A,$ where e is the identity element of G .

This action is also referred to as an action on A by left, or a **left action of G on A** .

Definition 1.2: Right Group Action

We can also define an action of G on A from the right by considering a mapping

$$* : A \times G \longrightarrow A, \quad (a, g) \longmapsto a * g$$

satisfying:

- (i) $(a * g_1) * g_2 = a * (g_1 g_2) \quad \forall g_1, g_2 \in G, \forall a \in A,$
- (ii) $a * e = a \quad \forall a \in A.$

1.1 Fundamental Examples

Example 1.3: Trivial G -Action

Let G be any group and $A \neq \emptyset$. Define

$$* : G \times A \longrightarrow A \quad \text{by} \quad g * a = a \quad \forall g \in G, \forall a \in A.$$

Let $g_1, g_2 \in G$ and $a \in A$. Then:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * a = a, \\ (g_1 g_2) * a &= a. \end{aligned}$$

Thus $g_1 * (g_2 * a) = (g_1 g_2) * a$, and clearly $e * a = a$. This shows that $*$ is a group action of G on A , known as the **trivial G -action**.

Example 1.4: Left Regular Action / Translation by Left

Take $A = G$. Define

$$* : G \times A \longrightarrow A \quad \text{by} \quad g * a = ga \in A \quad \forall g \in G, \forall a \in A = G.$$

Verification of axioms:

(i) For $g_1, g_2 \in G$ and $a \in A$:

$$g_1 * (g_2 * a) = g_1 * (g_2 a) = g_1(g_2 a) = (g_1 g_2) a = (g_1 g_2) * a.$$

(ii) $e * a = ea = a$.

Therefore, $*$ is a group action of G on $A (= G)$. This action of G on itself is known as the **action by left** or **translation by left**.

Example 1.5: Action by Right Translation

Consider again $A = G$ and a mapping $* : G \times A \rightarrow A$.

- If we define $g * a = ag$, then for $g_1, g_2 \in G$ and $a \in A$:

$$g_1 * (g_2 * a) = g_1 * (ag_2) = (ag_2)g_1 = a(g_2 g_1) \neq a(g_1 g_2) = (g_1 g_2) * a.$$

So this does *not* define a left group action in general.

- However, if we define $* : G \times A \rightarrow A$ as

$$g * a = ag^{-1} \quad \forall g \in G, \forall a \in A = G,$$

then for all $g_1, g_2 \in G$ and $a \in A$:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (ag_2^{-1}) \\ &= (ag_2^{-1})g_1^{-1} \\ &= a(g_2^{-1}g_1^{-1}) \\ &= a(g_1 g_2)^{-1} \\ &= (g_1 g_2) * a, \end{aligned}$$

and

$$e * a = ae^{-1} = a.$$

Therefore, $*$ is a group action, referred to as the **action of G by right translation** (or right multiplication). This is also known as the **regular action of G on itself**.

2 Actions by Conjugation

Example 2.1: Action by Conjugation on G

Let G be a group. Take $A = G$. Define

$$* : G \times A \longrightarrow A \quad \text{by} \quad g * a = gag^{-1} \quad \forall g \in G, \forall a \in A.$$

Then $*$ is a group action on G , known as the **action by conjugation**.

Verification of axioms. Take $g_1, g_2 \in G$ and $a \in A (= G)$:

(i)

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (g_2 a g_2^{-1}) \\ &= g_1 g_2 a g_2^{-1} g_1^{-1} \\ &= g_1 g_2 a (g_1 g_2)^{-1} \\ &= (g_1 g_2) * a. \end{aligned}$$

(ii)

$$e * a = eae^{-1} = a.$$

Therefore, the axioms hold and $*$ is a valid group action. ■

Example 2.2: Action by Conjugation on Cosets

Let G be a group and $H \trianglelefteq G$ be a normal subgroup. Take

$$A = G/H = \{aH \mid a \in G\},$$

the set of all left cosets of H in G . Define

$$* : G \times A \longrightarrow A \quad \text{by} \quad g * (xH) = gxg^{-1}H \quad \forall g \in G, \forall xH \in A.$$

Verification of axioms. Take $g_1, g_2 \in G$ and $xH \in A$:

(i)

$$\begin{aligned} g_1 * (g_2 * xH) &= g_1 * (g_2 x g_2^{-1} H) \\ &= g_1 g_2 x g_2^{-1} g_1^{-1} H \\ &= g_1 g_2 x (g_1 g_2)^{-1} H \\ &= (g_1 g_2) * xH. \end{aligned}$$

(ii)

$$e * xH = exe^{-1}H = xH.$$

Therefore, $*$ is a group action. ■

Example 2.3: Action by Conjugation on Subgroups

Let G be a group and let A be the set of all subgroups of G . Define

$$* : G \times A \longrightarrow A \quad \text{by} \quad g * H = gHg^{-1} \quad \forall g \in G, \forall H \in A.$$

Then $*$ is also a group action.

Verification of axioms. For $g_1, g_2 \in G$ and $H \in A$:

(i)

$$\begin{aligned} g_1 * (g_2 * H) &= g_1 * (g_2 H g_2^{-1}) \\ &= g_1 g_2 H g_2^{-1} g_1^{-1} \\ &= g_1 g_2 H (g_1 g_2)^{-1} \\ &= (g_1 g_2) * H. \end{aligned}$$

(ii)

$$e * H = e H e^{-1} = H.$$

Therefore, $*$ is a group action on the set of subgroups. ■

Remark

If $* : G \times A \rightarrow A$ is an action, then because $*$ is a map, if $x, y \in A$ such that $x = y$, then for all $g \in G$:

$$(g, x) = (g, y) \implies g * x = g * y.$$

3 Group Actions and Permutation Representations

Theorem 3.1

Let G be any group and A be a non-empty set. Then any homomorphism from G to $\text{Sym}(A)$, the symmetric group on A , defines an action of G on A . Conversely, every action of G on A induces a homomorphism from G to $\text{Sym}(A)$.

Proof. We establish the theorem in two parts:

Part 1: Homomorphism induces an action. Let $F : G \rightarrow \text{Sym}(A)$ be a group homomorphism. For each $g \in G$, denote $F(g) = \sigma_g$, where $\sigma_g : A \rightarrow A$ is a bijective map. Since F is a homomorphism, for any $g_1, g_2 \in G$:

$$F(g_1 g_2) = F(g_1) \circ F(g_2) \implies \sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}.$$

Define $* : G \times A \rightarrow A$ by:

$$g * a = \sigma_g(a) \quad \forall g \in G, \forall a \in A.$$

Then $*$ is a map. For any $g_1, g_2 \in G$ and $a \in A$:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (\sigma_{g_2}(a)) \\ &= \sigma_{g_1}(\sigma_{g_2}(a)) \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(a) \\ &= \sigma_{g_1 g_2}(a) \\ &= (g_1 g_2) * a. \end{aligned}$$

Furthermore, since F is a homomorphism, $F(e) = \sigma_e = I_A$, where I_A is the identity permutation on A . Thus:

$$e * a = \sigma_e(a) = I_A(a) = a.$$

Therefore, $*$ is a group action of G on A .

Part 2: Action induces a homomorphism. Conversely, suppose $*$: $G \times A \rightarrow A$ is a group action. Define $F : G \rightarrow \text{Sym}(A)$ by setting $F(g) = \sigma_g$ for each $g \in G$, where $\sigma_g : A \rightarrow A$ is defined as:

$$\sigma_g(a) = g * a \quad \forall a \in A.$$

First, we show that σ_g is a bijective map from A to A :

- **One-to-one (Injective):** Let $a_1, a_2 \in A$ such that $\sigma_g(a_1) = \sigma_g(a_2)$. Then:

$$\begin{aligned} g * a_1 = g * a_2 &\implies g^{-1} * (g * a_1) = g^{-1} * (g * a_2) \\ &\implies (g^{-1}g) * a_1 = (g^{-1}g) * a_2 \quad (\text{as } * \text{ is an action}) \\ &\implies e * a_1 = e * a_2 \\ &\implies a_1 = a_2. \end{aligned}$$

Hence, σ_g is one-to-one.

- **Onto (Surjective):** Let $a \in A$. Then $g^{-1} * a \in A$, and:

$$\sigma_g(g^{-1} * a) = g * (g^{-1} * a) = (gg^{-1}) * a = e * a = a.$$

Hence, σ_g is onto.

Thus σ_g is bijective, meaning $\sigma_g \in \text{Sym}(A)$, and F is a well-defined map.

Next, we show that F is a group homomorphism. For any $g_1, g_2 \in G$ and $a \in A$:

$$\begin{aligned} \sigma_{g_1 g_2}(a) &= (g_1 g_2) * a \\ &= g_1 * (g_2 * a) \\ &= g_1 * \sigma_{g_2}(a) \\ &= \sigma_{g_1}(\sigma_{g_2}(a)) \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(a). \end{aligned}$$

Therefore, $\sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}$, which implies:

$$F(g_1 g_2) = F(g_1) \circ F(g_2).$$

Hence, F is a group homomorphism. ■

Remark

For any group action of a group G on a set A , the induced homomorphism

$$F : G \longrightarrow \text{Sym}(A)$$

is called the **associated permutation representation** of the given action.

Corollary 3.2: Cayley's Theorem

Every group is isomorphic to a subgroup of its permutation group (symmetric group).

Proof. Let G be any group. We know that G acts on itself ($A = G$) under the left regular action:

$$* : G \times A \longrightarrow A, \quad g * a = ga \quad \forall g \in G, \forall a \in A = G.$$

By the previous theorem, the corresponding permutation representation is the homomorphism:

$$F : G \longrightarrow \text{Sym}(G), \quad F(g) = \sigma_g, \quad \text{where } \sigma_g(a) = ga.$$

We show that F is one-to-one:

$$\begin{aligned} F(g_1) = F(g_2) &\implies \sigma_{g_1} = \sigma_{g_2} \\ &\implies \sigma_{g_1}(a) = \sigma_{g_2}(a) \quad \forall a \in G \\ &\implies g_1 a = g_2 a \quad \forall a \in G. \end{aligned}$$

Choosing $a = e$:

$$g_1 e = g_2 e \implies g_1 = g_2.$$

Therefore, F is one-to-one, which means:

$$\ker F = \{e\}.$$

By the Fundamental Theorem of Group Homomorphisms (First Isomorphism Theorem):

$$\frac{G}{\ker F} \cong F(G) \implies \frac{G}{\{e\}} \cong F(G) \implies G \cong F(G).$$

Since $F(G) \leq \text{Sym}(G)$, every group G is isomorphic to a subgroup of $\text{Sym}(G)$. ■

4 Kernel of an Action

Definition 4.1: Kernel of an Action

Let $* : G \times A \rightarrow A$ be a group action. The **kernel** of $*$ is defined as:

$$\ker * = \{g \in G \mid g * a = a, \forall a \in A\}.$$

In other words, $\ker *$ consists of all elements of G that fix every element of A .

Theorem 4.2

Let $* : G \times A \rightarrow A$ be a group action. Then:

$$\ker * \leq G.$$

Proof. We check the subgroup criteria:

- (i) **Non-emptiness:** Since $e * a = a$ for all $a \in A$, we have $e \in \ker *$, so $\ker * \neq \emptyset$.
(ii) **Closure under multiplication:** Let $g_1, g_2 \in \ker *$. Then:

$$g_1 * a = a \quad \text{and} \quad g_2 * a = a \quad \forall a \in A.$$

Then for any $a \in A$:

$$(g_1 g_2) * a = g_1 * (g_2 * a) = g_1 * a = a.$$

Thus $g_1 g_2 \in \ker *$.

- (iii) **Closure under inversion:** Let $g_1 \in \ker *$, so $g_1 * a = a$ for all $a \in A$. Then:

$$\begin{aligned} g_1^{-1} * (g_1 * a) &= g_1^{-1} * a \implies (g_1^{-1} g_1) * a = g_1^{-1} * a \\ &\implies e * a = g_1^{-1} * a \\ &\implies a = g_1^{-1} * a \quad \forall a \in A. \end{aligned}$$

Thus $g_1^{-1} \in \ker *$.

Therefore, $\ker * \leq G$. ■

Theorem 4.3

If $F : G \rightarrow \text{Sym}(A)$ is the permutation representation corresponding to the group action $*$, then:

$$\ker F = \ker *.$$

Proof. Recall that $F(g) = \sigma_g$ where $\sigma_g(a) = g * a$ for all $a \in A$. By the definition of the kernel of a group homomorphism:

$$\begin{aligned} \ker F &= \{g \in G \mid F(g) = I_A\} \\ &= \{g \in G \mid \sigma_g = \sigma_e\} \\ &= \{g \in G \mid \sigma_g(a) = \sigma_e(a) \quad \forall a \in A\} \\ &= \{g \in G \mid g * a = e * a \quad \forall a \in A\} \\ &= \{g \in G \mid g * a = a \quad \forall a \in A\} \\ &= \ker *. \end{aligned}$$

Hence, $\ker F = \ker *$. ■

5 Orbits and Stabilizers

Definition 5.1: Stabilizer

Let $*$ be a group action of a group G on a set A , and let $a \in A$. The set

$$G_a = \{g \in G \mid g * a = a\}$$

is called the **stabilizer** (or isotropy group) of a in G .

Theorem 5.2

For any $a \in A$, the stabilizer G_a is a subgroup of G :

$$G_a \leq G.$$

Proof. We check the subgroup criteria:

(i) Since $e * a = a$, we have $e \in G_a$, so $G_a \neq \emptyset$.

(ii) Let $g_1, g_2 \in G_a$. Then:

$$g_1 * a = a \quad \text{and} \quad g_2 * a = a.$$

Now, using the fact that $*$ is a group action:

$$(g_1 g_2) * a = g_1 * (g_2 * a) = g_1 * a = a.$$

Therefore:

$$g_1 g_2 \in G_a \longrightarrow (1)$$

(iii) Also, since $g_1 * a = a$:

$$\begin{aligned} g_1^{-1} * (g_1 * a) &= g_1^{-1} * a \implies (g_1^{-1} g_1) * a = g_1^{-1} * a \\ &\implies e * a = g_1^{-1} * a \\ &\implies g_1^{-1} * a = a. \end{aligned}$$

Therefore:

$$g_1^{-1} \in G_a \longrightarrow (2)$$

From (1) and (2), we conclude that $G_a \leq G$. ■

Definition 5.3: Orbit

Let $*$ be a group action of G on A , and let $a \in A$. The **orbit** of a under G is the set:

$$G^a = \{g * a \mid g \in G\} \subseteq A.$$

Theorem 5.4: Orbit-Stabilizer Theorem

Let $*$ be an action of a group G on a set A , and let $a \in A$. There is a bijection between the orbit G^a and the set of left cosets of the stabilizer, G/G_a .

Proof. Define the map

$$\phi : G^a \longrightarrow G/G_a \quad \text{by} \quad \phi(g * a) = gG_a.$$

We show that ϕ is well-defined, injective, and surjective:

• **Well-defined:** Let $g_1 * a = g_2 * a$. Then:

$$\begin{aligned} g_2^{-1} * (g_1 * a) &= g_2^{-1} * (g_2 * a) \implies (g_2^{-1} g_1) * a = (g_2^{-1} g_2) * a \\ &\implies (g_2^{-1} g_1) * a = e * a = a \\ &\implies g_2^{-1} g_1 \in G_a \\ &\implies g_1 G_a = g_2 G_a \\ &\implies \phi(g_1 * a) = \phi(g_2 * a). \end{aligned}$$

Therefore, ϕ is well-defined.

- **One-to-one (Injective):** Let $\phi(g_1 * a) = \phi(g_2 * a)$. Then:

$$\begin{aligned}
 g_1 G_a = g_2 G_a &\implies g_2^{-1} g_1 \in G_a \\
 &\implies (g_2^{-1} g_1) * a = a \\
 &\implies g_2 * ((g_2^{-1} g_1) * a) = g_2 * a \\
 &\implies (g_2 g_2^{-1} g_1) * a = g_2 * a \\
 &\implies g_1 * a = g_2 * a.
 \end{aligned}$$

Therefore, ϕ is one-to-one.

- **Onto (Surjective):** Let $t \in G/G_a$. Then $t = gG_a$ for some $g \in G$. For this element g , $g * a \in G^a$ satisfies:

$$\phi(g * a) = gG_a = t.$$

Therefore, ϕ is onto.

Thus, ϕ is a bijection between G^a and G/G_a . ■

Note

If G is a finite group, then:

$$|G^a| = |G/G_a| = \frac{|G|}{|G_a|},$$

which yields the fundamental relation:

$$|G| = |G^a| \cdot |G_a|.$$

6 Orbits as Equivalence Classes and Conjugacy Classes

6.1 Equivalence Relation Induced by an Action

Theorem 6.1

Let $*$ be a group action of a group G on a set A , $* : G \times A \rightarrow A$. Define a relation $\sim$ on A by:

$$a \sim b \iff a = g * b \text{ for some } g \in G.$$

Then $\sim$ is an equivalence relation on A .

Proof. We verify the three equivalence relation properties:

- (i) **Reflexive:** For all $a \in A$, since $e \in G$ is the identity:

$$a = e * a \implies a \sim a.$$

Therefore, $\sim$ is reflexive.

- (ii) **Symmetric:** Let $a, b \in A$ such that $a \sim b$. Then there exists $g \in G$ such that:

$$a = g * b.$$

Applying $g^{-1} \in G$ to both sides:

$$\begin{aligned}
 g^{-1} * a &= g^{-1} * (g * b) \\
 &= (g^{-1}g) * b \\
 &= e * b = b.
 \end{aligned}$$

Since $g^{-1} \in G$, we have $b \sim a$. Therefore, $\sim$ is symmetric.

- (iii) **Transitive:** Let $a, b, c \in A$ such that $a \sim b$ and $b \sim c$. Then there exist $g_1, g_2 \in G$ such that:

$$a = g_1 * b \quad \text{and} \quad b = g_2 * c.$$

Substituting b into the expression for a :

$$a = g_1 * (g_2 * c) = (g_1 g_2) * c.$$

Since $g_1 g_2 \in G$, we have $a \sim c$. Therefore, $\sim$ is transitive.

Hence, $\sim$ is an equivalence relation on A . ■

6.2 Partition of the Set into Orbits

Since $\sim$ is an equivalence relation on A , it partitions the set A into its equivalence classes. For any $a \in A$, the equivalence class of a is:

$$\begin{aligned} \text{cl}(a) &= \{b \in A \mid a \sim b\} \\ &= \{b \in A \mid b = g * a \text{ for some } g \in G\} \\ &= \{g * a \mid g \in G\} \\ &= G^a \quad (\text{the orbit of } a \text{ under } G). \end{aligned}$$

Consequently, the set A decomposes as a disjoint union of its orbits:

$$A = \bigcup_{a \in A} \text{cl}(a) = \bigcup_{a \in A} G^a.$$

6.3 Conjugacy Classes

If G acts on itself ($A = G$) by conjugation, where $g * a = gag^{-1}$:

- The stabilizer of a is the **normalizer** (or centralizer) of a :

$$G_a = N(a) = \{g \in G \mid gag^{-1} = a\} = \{g \in G \mid ga = ag\}.$$

- The orbit of a is the **conjugacy class** of a :

$$G^a = \text{cl}(a) = \{g * a \mid g \in G\} = \{gag^{-1} \mid g \in G\}.$$

Exercise 6.2

Let G be a group and $a, b \in G$. Then a and b are conjugate if there exists $c \in G$ such that $b = cac^{-1}$. Prove that this conjugacy relation $\sim$ is an equivalence relation on G .

Solution. We check the three properties:

- (i) **Reflexive:** Let $a \in G$. Since $e \in G$, we have:

$$a = eae^{-1}.$$

Therefore, $a \sim a$, so $\sim$ is reflexive.

(ii) **Symmetric:** Let $a, b \in G$ such that $a \sim b$. Then there exists $c \in G$ such that:

$$a = cbc^{-1}.$$

Multiplying on the left by c^{-1} and on the right by c :

$$\begin{aligned} c^{-1}a &= c^{-1}(cbc^{-1}) = bc^{-1}, \\ c^{-1}ac &= bc^{-1}c = be = b. \end{aligned}$$

Writing $c = (c^{-1})^{-1}$, this gives:

$$c^{-1}a(c^{-1})^{-1} = b.$$

Since $c^{-1} \in G$, this shows $b \sim a$. Therefore, $\sim$ is symmetric.

(iii) **Transitive:** Let $a, b, c \in G$ such that $a \sim b$ and $b \sim c$. Then there exist $d_1, d_2 \in G$ such that:

$$a = d_1bd_1^{-1} \quad \text{and} \quad b = d_2cd_2^{-1}.$$

Substituting b into the first equation:

$$\begin{aligned} a &= d_1(d_2cd_2^{-1})d_1^{-1} \\ &= d_1d_2cd_2^{-1}d_1^{-1} \quad \{d_2^{-1}d_1^{-1} = (d_1d_2)^{-1}\} \\ &= (d_1d_2)c(d_1d_2)^{-1}. \end{aligned}$$

Since $d_1, d_2 \in G \implies d_1d_2 \in G$, it follows that $a \sim c$.

Therefore, $\sim$ is transitive, and hence an equivalence relation on G . ■

7 The Class Equation and Center of p -Groups

7.1 Orbit-Stabilizer Applied to Conjugation

By the Orbit-Stabilizer Theorem, for any $a \in G$, there exists a bijection between the conjugacy class $\text{cl}(a)$ and the set of cosets $G/N(a)$, for all $a \in G$. Consequently:

$$G = \bigcup_{a \in G} \text{cl}(a).$$

If G is a finite group, then:

$$\begin{aligned} |\text{cl}(a)| &= \left| \frac{G}{N(a)} \right| = \frac{|G|}{|N(a)|}, \\ |G| &= \sum_{a \in G} |\text{cl}(a)|. \end{aligned}$$

Proposition 7.1

For any element $a \in G$:

$$\text{cl}(a) = \{a\} \iff a \in Z(G),$$

where $Z(G) = \{z \in G \mid zg = gz \forall g \in G\}$ is the center of G .

Proof. By definition of the conjugacy class:

$$\begin{aligned} \text{cl}(a) = \{a\} &\iff gag^{-1} = a \quad \forall g \in G \\ &\iff ga = ag \quad \forall g \in G \\ &\iff a \in Z(G). \end{aligned}$$
■

7.2 The Class Equation

Separating the central elements (for which $|\text{cl}(a)| = 1$) from the non-central elements yields the **Class Equation**:

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right| \rightarrow (1)$$

Theorem 7.2

If $|G| = p^n$, where p is a prime and $n \geq 1$, then:

$$|Z(G)| \geq p.$$

In other words, every non-trivial finite p -group has a non-trivial center.

Proof. Let $|Z(G)| = r$. By the class equation (1):

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right|.$$

For any $a \notin Z(G)$, $N(a) \neq G$, meaning $N(a)$ is a proper subgroup of G . By Lagrange's Theorem, $|N(a)|$ divides $|G| = p^n$, so:

$$|N(a)| = p^{\alpha_a} \quad \text{with } \alpha_a < n.$$

Then:

$$\left| \frac{G}{N(a)} \right| = \frac{|G|}{|N(a)|} = \frac{p^n}{p^{\alpha_a}} = p^{n-\alpha_a}, \quad \text{where } n - \alpha_a > 0.$$

This implies:

$$p \mid p^{n-\alpha_a} \implies p \mid \left| \frac{G}{N(a)} \right| \implies p \mid \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right|.$$

Since $|G| = p^n$, we also have $p \mid |G|$. Rewriting the class equation:

$$|Z(G)| = |G| - \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right|.$$

Since p divides both terms on the right-hand side:

$$p \mid |Z(G)| \implies p \mid r.$$

Since the identity element $e \in Z(G)$, the center is never empty, so $|Z(G)| \geq 1$. Because p divides r , we must have:

$$r \geq p \implies |Z(G)| \geq p.$$

■

Theorem 7.3

A group of order p^2 , where p is prime, is abelian.

Proof. Let $|G| = p^2$. By the previous theorem:

$$|Z(G)| \geq p.$$

By Lagrange's Theorem, $|Z(G)|$ must divide $|G| = p^2$. Therefore, the only possibilities are:

$$|Z(G)| = p \quad \text{or} \quad |Z(G)| = p^2.$$

If $|Z(G)| = p$, then:

$$\frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p.$$

Since any group of prime order is cyclic, the quotient group $G/Z(G)$ is cyclic. But it is a standard result that:

$$G/Z(G) \text{ is cyclic} \implies G \text{ is abelian.}$$

If G is abelian, every element commutes with all other elements, so:

$$G = Z(G) \implies |G| = |Z(G)| = p^2,$$

which contradicts the assumption that $|Z(G)| = p$.

Therefore:

$$|Z(G)| \neq p \quad (\text{contradiction}).$$

So we must have:

$$|Z(G)| = p^2 = |G| \implies G = Z(G).$$

Hence, G is abelian. ■

Remark

If $H \leq G$ such that $H \subseteq Z(G)$, then $H \trianglelefteq G$ (any subgroup contained in the center is normal in G).

8 Conjugacy Classes in the Symmetric Group S_n

Theorem 8.1: Number of Conjugates in S_n

Let σ be a permutation in S_n and let $m_1, m_2, \dots, m_s$ be the distinct integers which appear in the cycle type of σ (including 1-cycles). For each $i \in \{1, 2, \dots, s\}$, assume σ has k_i cycles of length m_i so that:

$$\sum_{i=1}^s k_i m_i = n.$$

Then the number of conjugates of σ in S_n (i.e., the size of the conjugacy class $|\text{cl}(\sigma)|$) is:

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_s! m_s^{k_s})}.$$

8.1 Examples in S_4 and S_5 **Example 8.2: Conjugacy Classes in S_4**

(a) **Number of cycles of length 2 in S_4 :** A transposition in S_4 has cycle decomposition of the form $(a\ b)(c)(d)$. Here:

- $k_1 = 1$ cycle of length $m_1 = 2$,
- $k_2 = 2$ cycles of length $m_2 = 1$.

Applying the formula:

$$\frac{4!}{(1! \cdot 2^1)(2! \cdot 1^2)} = \frac{4 \times 3 \times 2 \times 1}{2 \times 2} = 6.$$

Explicitly, the conjugacy class of $(1\ 2)$ consists of all 6 transpositions:

$$\text{cl}((1\ 2)) = \{(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4)\}.$$

(b) **Cycles of length 3 in S_4 :** A 3-cycle has cycle type $(a\ b\ c)(d)$, so $k_1 = 1, m_1 = 3$ and $k_2 = 1, m_2 = 1$:

$$|\text{cl}((1\ 2\ 3))| = \frac{4!}{(1! \cdot 3^1)(1! \cdot 1^1)} = \frac{24}{3} = 8.$$

Example 8.3: Conjugacy Classes in S_5

(a) **Cycles of length 2 in S_5 :** A single transposition has cycle structure $(a\ b)(c)(d)(e)$, so $k_1 = 1, m_1 = 2$ and $k_2 = 3, m_2 = 1$:

$$\text{No. of cycles of length 2} = \frac{5!}{(3! \cdot 1^3)(1! \cdot 2^1)} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times 2} = \frac{20}{2} = 10.$$

(b) **Permutations with cycle type $(2, 3)$ in S_5 :** A product of a disjoint transposition and 3-cycle has form $(a\ b)(c\ d\ e)$, so $k_1 = 1, m_1 = 2$ and $k_2 = 1, m_2 = 3$:

$$\frac{5!}{(1! \cdot 2^1)(1! \cdot 3^1)} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 3} = 20.$$

Part II: Introduction to Ring Theory

9 Introduction to Ring Theory: Definitions and Axioms

Definition 9.1: Ring

A algebraic system $(R, +, \cdot)$ consisting of a non-empty set R equipped with two internal binary operations, denoted $+$ (addition) and $\cdot$ (multiplication), is called a **ring** if it satisfies the following three conditions:

1. $(R, +)$ is an **abelian group**:

- *Closure*: $a + b \in R$ for all $a, b \in R$.
- *Associativity*: $(a + b) + c = a + (b + c)$ for all $a, b, c \in R$.
- *Identity*: There exists an element $0 \in R$ such that $a + 0 = 0 + a = a$ for all $a \in R$.
- *Inverse*: For each $a \in R$, there exists $-a \in R$ such that $a + (-a) = (-a) + a = 0$.
- *Commutativity*: $a + b = b + a$ for all $a, b \in R$.

2. $(R, \cdot)$ is a **semi-group**:

- *Closure*: $a \cdot b \in R$ for all $a, b \in R$.
- *Associativity*: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$.

3. **Distributive Laws hold**: Multiplication distributes over addition from both left and right:

$$\begin{aligned} a \cdot (b + c) &= a \cdot b + a \cdot c \quad \forall a, b, c \in R, \\ (a + b) \cdot c &= a \cdot c + b \cdot c \quad \forall a, b, c \in R. \end{aligned}$$

9.1 Terminology and Ring Properties

1. The operation $+$ is usually referred to as **addition**, and $\cdot$ is usually referred to as **multiplication**.
2. The identity element of the abelian group $(R, +)$ is called the **zero element** of the ring and is denoted as 0.
3. The identity element with respect to multiplication $\cdot$ (if it exists) is known as the **identity element of the ring** or **unity**, and is denoted as 1.
4. If the multiplication operation $\cdot$ is commutative, i.e., $a \cdot b = b \cdot a$ for all $a, b \in R$, then R is called a **commutative ring**.
5. If the multiplicative identity exists in R , then R is called a **ring with identity** or **ring with unity**.

10 Examples of Rings

10.1 Commutative Rings with Unity

Example 10.1: Standard Number Systems

The sets of numbers with standard addition and multiplication:

$$(\mathbb{Z}, +, \cdot), \quad (\mathbb{Q}, +, \cdot), \quad (\mathbb{R}, +, \cdot), \quad (\mathbb{C}, +, \cdot)$$

are all commutative rings with unity (1).

Example 10.2: Integers Modulo n

For any positive integer $n \geq 2$, the set of congruence classes:

$$(\mathbb{Z}_n, +, \cdot)$$

under addition and multiplication modulo n forms a commutative ring with unity ($[1]$).

Example 10.3: Quadratic Integer Ring

Let p be a prime. Define the set:

$$\mathbb{Z}[\sqrt{p}] = \{a + b\sqrt{p} \mid a, b \in \mathbb{Z}\}.$$

Then $\mathbb{Z}[\sqrt{p}]$ is a commutative ring with unity ($1 = 1 + 0\sqrt{p}$).

Example 10.4: Ring of Gaussian Integers

The set of Gaussian integers:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}, i^2 = -1\}$$

is a commutative ring with unity ($1 = 1 + 0i$).

10.2 Non-Commutative Rings with Unity

Example 10.5: Real Matrix Ring $M_n(\mathbb{R})$

Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries:

- (i) $(M_n(\mathbb{R}), +)$ is an abelian group with zero element the zero matrix $O_{n \times n}$.
- (ii) $(M_n(\mathbb{R}), \cdot)$ is a semi-group under matrix multiplication (associativity of matrix multiplication).
- (iii) The distributive laws hold:

$$\begin{aligned} A \cdot (B + C) &= AB + AC, \\ (A + B) \cdot C &= AC + BC. \end{aligned}$$

- (iv) The identity matrix I_n satisfies $AI_n = I_nA = A$ for all $A \in M_n(\mathbb{R})$.
- (v) Matrix multiplication is not commutative for $n \geq 2$ (in general, $AB \neq BA$).

Therefore, $(M_n(\mathbb{R}), +, \cdot)$ is a **non-commutative ring with unity**.

Example 10.6: Integer Matrix Ring $M_n(\mathbb{Z})$

Similarly, let $M_n(\mathbb{Z})$ denote the set of all $n \times n$ matrices with integer entries. Then $(M_n(\mathbb{Z}), +, \cdot)$ is also a **non-commutative ring with unity**.

11 Special Matrix Ring: Identity and Inverse Computations

Example 11.1: A Commutative Matrix Ring with Non-Standard Unity

Consider the set:

$$A = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \mid x \in \mathbb{R} \right\}$$

under standard matrix addition and multiplication.

11.1 Verification of Ring Axioms

1. **$(A, +)$ is an abelian group:** For any $X = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$ and $Y = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \in A$:

$$X + Y = \begin{bmatrix} x + y & x + y \\ x + y & x + y \end{bmatrix} \in A.$$

Addition is associative and commutative since addition in $\mathbb{R}$ is. The zero element is:

$$\mathbf{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in A,$$

and the additive inverse of X is $-X = \begin{bmatrix} -x & -x \\ -x & -x \end{bmatrix} \in A$.

2. **Commutativity of multiplication in A :** Let $X, Y \in A$ with $X = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$ and $Y = \begin{bmatrix} y & y \\ y & y \end{bmatrix}$. Then:

$$\begin{aligned} \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} &= \begin{bmatrix} x \cdot y + x \cdot y & x \cdot y + x \cdot y \\ x \cdot y + x \cdot y & x \cdot y + x \cdot y \end{bmatrix} \\ &= \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} \\ &= \begin{bmatrix} 2yx & 2yx \\ 2yx & 2yx \end{bmatrix} \\ &= \begin{bmatrix} y & y \\ y & y \end{bmatrix} \begin{bmatrix} x & x \\ x & x \end{bmatrix}. \end{aligned}$$

Therefore, multiplication is commutative in A .

3. **Distributivity:** Matrix multiplication is always distributive over matrix addition.

Thus, $(A, +, \cdot)$ is a **commutative ring**.

11.2 Finding the Unity Element

Let $E \in A$ be the multiplicative identity (unity) of A , so that:

$$E = \begin{bmatrix} e & e \\ e & e \end{bmatrix}.$$

Then E must satisfy $XE = X$ for all $X \in A$:

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} e & e \\ e & e \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix}.$$

Performing the multiplication:

$$\begin{bmatrix} 2xe & 2xe \\ 2xe & 2xe \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix}.$$

Equating corresponding entries:

$$2xe = x \implies 2e = 1 \implies e = \frac{1}{2}.$$

Therefore, the identity element of the ring A is:

$$E = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}.$$

Remark

Notice that the unity element $E \in A$ is *not* the standard 2×2 identity matrix $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (which does not belong to A), yet it functions as the exact multiplicative identity on A .

11.3 Computation of Multiplicative Inverses

For $(A \setminus \{\mathbf{0}\}, \cdot)$ to form an abelian group (making A a field), each non-zero element must possess a multiplicative inverse. Let:

$$Y = X^{-1} = \begin{bmatrix} y & y \\ y & y \end{bmatrix}.$$

Then $XY = E$:

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}.$$

This gives:

$$\begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix},$$

which implies:

$$2xy = \frac{1}{2} \implies y = \frac{1}{4x} \quad (x \neq 0).$$

Therefore, for any non-zero element $X \in A$, its multiplicative inverse is:

$$X^{-1} = \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix}.$$

Consequently, $(A \setminus \{\mathbf{0}\}, \cdot)$ is an abelian group, and A is isomorphic as a field to $\mathbb{R}$.